

# À vous de jouer !

## Differential Fault Analysis

### Défi 4 : DFA au 16e tour sur une boîte $S$ – Solution

R1 $\mathrm{P}_1^{-1}\left(\Delta R_{16}\right) = \mathrm{S}_1\left(\mathrm{E}_1\left(L_{16}\right) \oplus K_{16,1}\right) \oplus \mathrm{S}_1\left(\mathrm{E}_1\left(L'_{16}\right) \oplus K_{16,1}\right)$

R2 $\mathtt{0xb} = \mathrm{S}_1\left(\mathtt{0x12} \oplus K_{16,1}\right) \oplus \mathrm{S}_1\left(\mathtt{0x1a} \oplus K_{16,1}\right) = \Delta^4 \mathrm{S}_1$

| $\Delta^4 \mathrm{S}_1(x)$ | $x$ | $\Delta^4 \mathrm{S}_1(x)$ | $x$ |
|---|---|---|---|
| 0x0 | - | 0x8 | - |
| 0x1 | - | 0x9 | 0x14, 0x1c, 0x32, 0x37 0x3a, 0x3f |
| 0x2 | - | 0xa | 0x17, 0x1f |
| 0x3 | 0x00, 0x01, 0x03, 0x05 0x08, 0x09, 0x0b, 0x0d 0x25, 0x2d, 0x34, 0x3c | 0xb | 0x04, 0x0c, 0x10, 0x18 0x21, 0x29, 0x33, 0x3b |
| 0x4 | - | 0xc | 0x12, 0x1a, 0x22, 0x26 0x2a, 0x2e, 0x36, 0x3e |
| 0x5 | 0x15, 0x1d, 0x27, 0x2f 0x31, 0x35, 0x39, 0x3d | 0xd | 0x11, 0x19 |
| 0x6 | 0x02, 0x06, 0x0a, 0x0e 0x16, 0x1e, 0x20, 0x28 | 0xe | 0x13, 0x1b |
| 0x7 | 0x24, 0x2c, 0x30, 0x38 | 0xf | 0x07, 0x0f, 0x23, 0x2b |

$\Rightarrow K_{16,1} \in \{\mathtt{0x04}, \mathtt{0x0c}, \mathtt{0x10}, \mathtt{0x18}, \mathtt{0x21}, \mathtt{0x29}, \mathtt{0x33}, \mathtt{0x3b}\}$